PROB. AND STAT. LAB FILE

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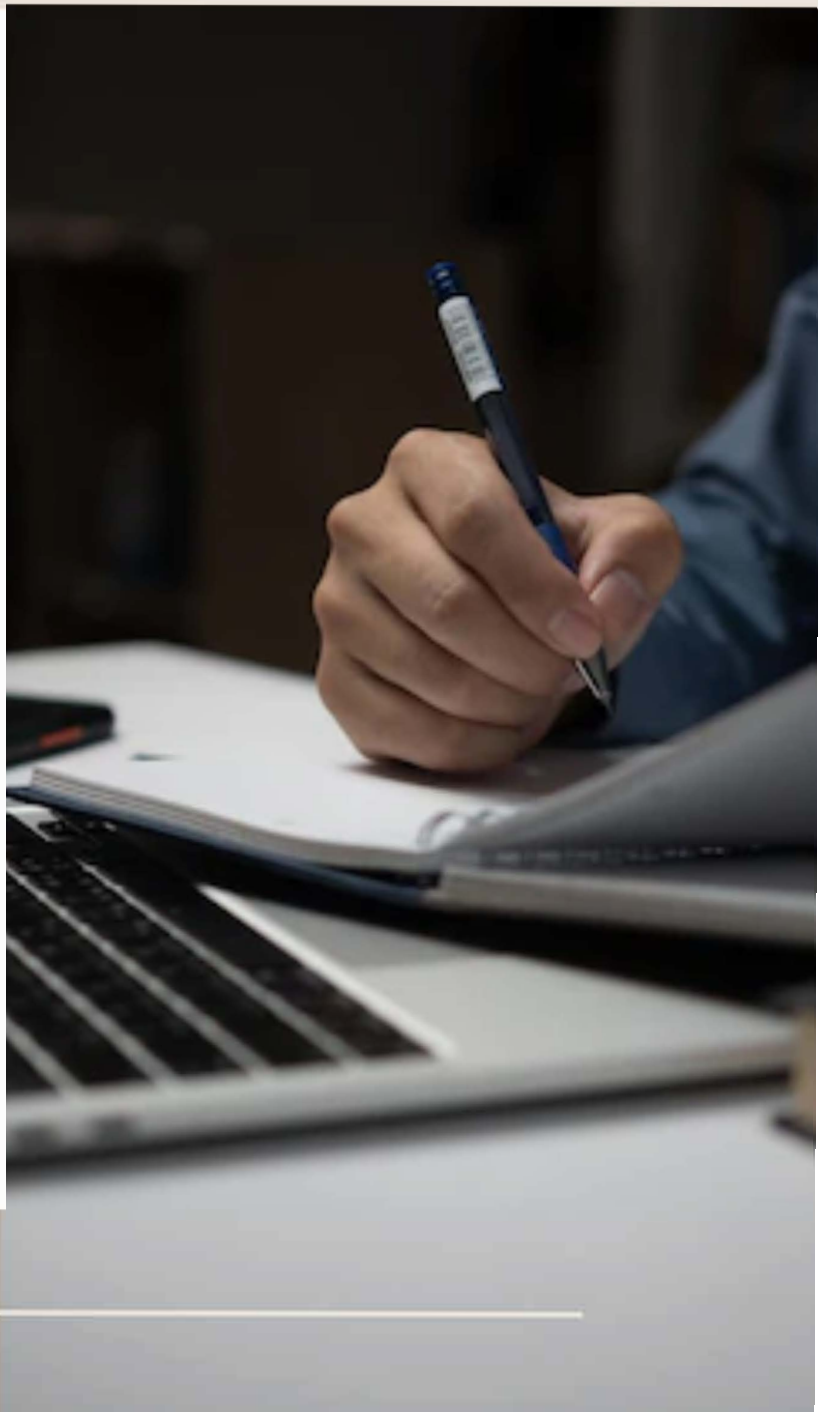


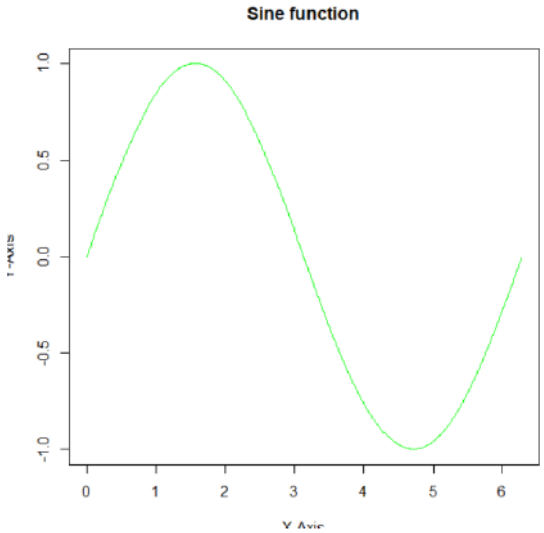
2025

ASSIGNMENT 1

Q1
```
seq(from = 1, to = 20, by = 1)
# OUTPUT:
# [1]  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20
```

Q2
```
X <- c(10,20,30,40,50,60)
max(X)
min(X)
# OUTPUT:
# max(X) -> [1] 60
# min(X) -> [1] 10
```

Q3
```
x <- seq(from = 0, to = 2*pi, by = 0.01)
y <- sin(x)
plot(x, y, type = "l", col = "green", xlab = "X-Axis", ylab = "Y-Axis", main = "Sine function")
# OUTPUT: (Sine function plot)
```



Q4
```
num <- as.integer(readline(prompt = "Enter a number: "))
if (num < 0) {
cat("Warning : Negative Number")
} else {
fact <- factorial(num)
cat("Factorial of", num, "is", fact, "\n")
}
# Example OUTPUT (when input 4):
# Factorial of 4 is 24
```

Q5
```
x <- c(1,2,3,4)
mean(x)
# OUTPUT:
# [1] 2.5
```

Q6
```
num1 <- as.integer(readline(prompt = "Enter a number: "))
x <- 1
y <- 1
cat(x, "\n")
cat(y, "\n")
if(num1 < 0){
print("Error , Negative Input")
} else {
```

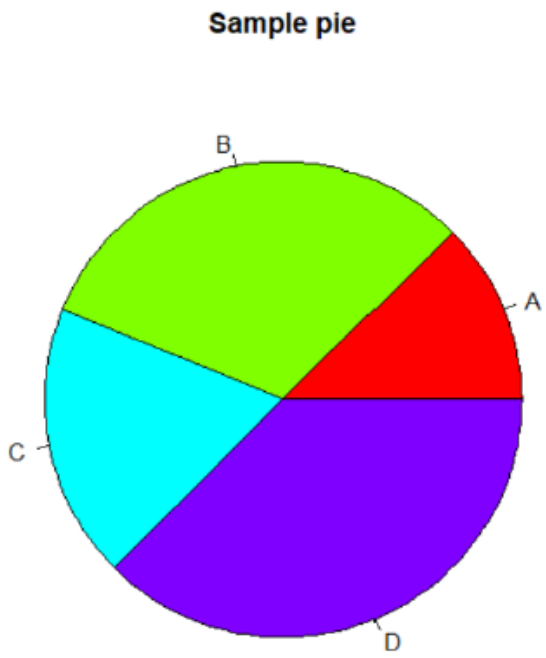
```
if(num1 >= 3){
  for(i in 3:num1){
    fib <- x + y
    x <- y
    y <- fib
    cat(fib, "\n")
  }
}
}
# OUTPUT: (Fibonacci sequence printed up to n terms)
```

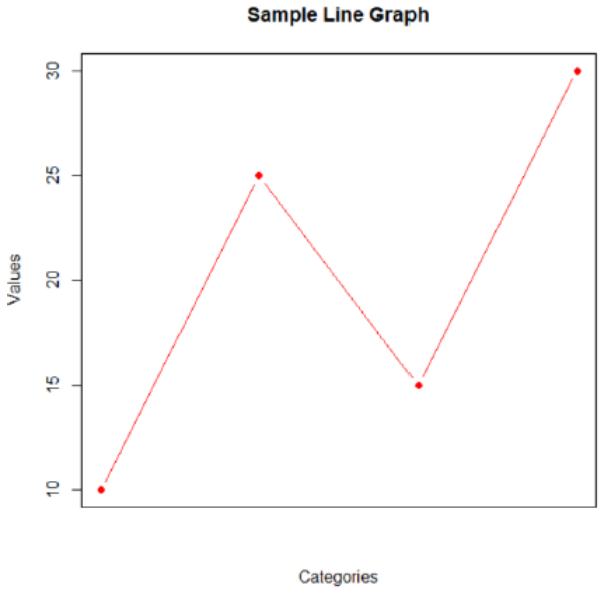
Q7

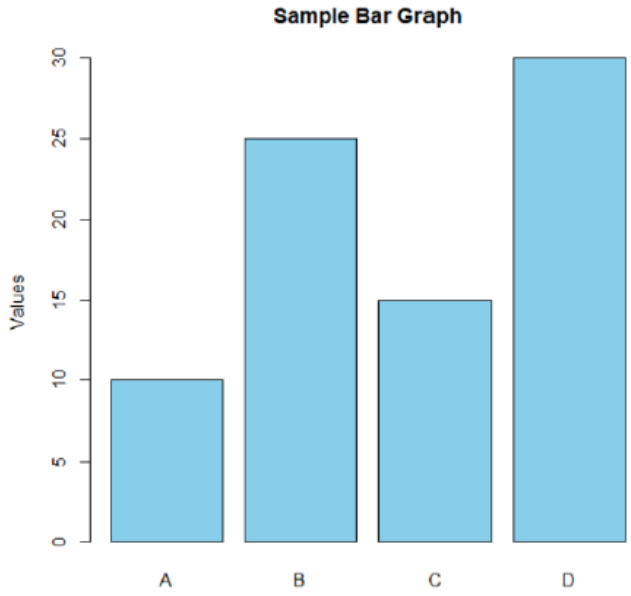
```
num1 <- as.numeric(readline(prompt = "Enter first number : "))
num2 <- as.numeric(readline(prompt = "Enter second number : "))
op <- readline(prompt = "Enter operator : ")
if(op == "+"){
result <- num1 + num2
} else if(op == "-"){
result <- num1 - num2
} else if(op == "*"){
result <- num1 * num2
} else if(op == "/"){
if(num2 == 0){
  result <- "Error : Division by zero"
} else {
  result <- num1 / num2
}
} else {
result <- "Invalid operator"
}
cat("Result : ", result, "\n")
# Example INPUT: 12, 3, /
# OUTPUT: Result : 4
```

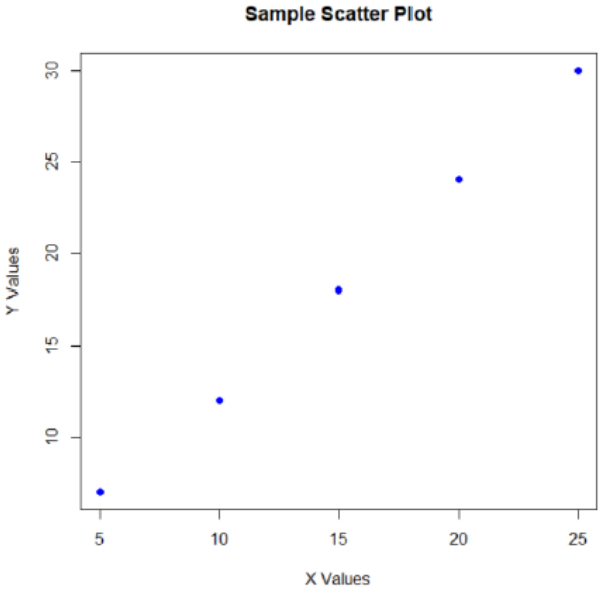
Q8

```
categories <- c("A","B","C","D")
values <- c(10,25,15,30)
pie(values, labels = categories, main = "Sample pie", col = rainbow(length(values)))
barplot(values, names.arg = categories, main = "Sample Bar Graph", col = "skyblue", ylab =
"Values")
plot(values, type = "b", main = "Sample Line Graph", xlab = "Categories", ylab = "Values", xaxt = "n",
col = "red", pch = 19)
# Sample scatter plot
x <- c(5, 10, 15, 20, 25)
y <- c(7, 12, 18, 24, 30)
plot(x, y, main = "Sample Scatter Plot", xlab = "X Values", ylab = "Y Values", pch = 19, col = "blue")
# OUTPUT: (Pie, bar, line, and scatter plots displayed)
```


Sample pie
B
A
C
D


Sample Line Graph
Values
10
15
20
25
30
Categories


Sample Bar Graph
Values
0
5
10
15
20
25
30
A
B
C
D


Sample Scatter Plot
Y Values
10
15
20
25
30
5
10
15
20
25
X Values

ASSIGNMENT 2

Q1

```
data(iris)
head(iris)
diff(range(iris$Sepal.Length))
diff(range(iris$Sepal.Width))
diff(range(iris$Petal.Length))
diff(range(iris$Petal.Width))
mean(iris$Petal.Width)
mean(iris$Sepal.Length)
mean(iris$Sepal.Width)
mean(iris$Petal.Length)
median(iris$Petal.Width)
median(iris$Sepal.Length)
median(iris$Sepal.Width)
median(iris$Petal.Length)
quantile(iris$Sepal.Length, 0.25)
quantile(iris$Sepal.Length, 0.75)
IQR(iris$Sepal.Length)
quantile(iris$Petal.Length, 0.25)
quantile(iris$Petal.Length, 0.75)
IQR(iris$Sepal.Length)
quantile(iris$Sepal.Width, 0.25)
quantile(iris$Sepal.Width, 0.75)
IQR(iris$Sepal.Length)
quantile(iris$Petal.Width, 0.25)
quantile(iris$Petal.Width, 0.75)
IQR(iris$Sepal.Length)
var(iris$Petal.Width)
var(iris$Sepal.Length)
var(iris$Sepal.Width)
var(iris$Petal.Length)
sd(iris$Petal.Width)
sd(iris$Sepal.Length)
sd(iris$Sepal.Width)
sd(iris$Petal.Length)
summary(iris)
# OUTPUT (excerpt):
# diff range Sepal.Length -> 3.6
# mean Sepal.Length -> 5.843333
# mean Sepal.Width -> 3.057333
# mean Petal.Length -> 3.758
# mean Petal.Width -> 1.199333
# summary shows min/1stQ/median/mean/3rdQ/max and species counts
```

Q2

```
mode <- function(x){
freq <- table(x)
mode_val <- names(freq)[freq == max(freq)]
if(is.numeric(x)){
  mode_val <- as.numeric(mode_val)
```

```
}
return(mode_val)
}
x <- c(1, 2, 2, 3, 4, 4, 4, 5)
mode(x)
# OUTPUT:
# [1] 4


Q3
birthday_same <- function(M, trials = 10000) {
matches <- 0
for (i in 1:trials) {
  birthdays <- sample(1:365, M, replace = TRUE)
  if (any(duplicated(birthdays))) {
    matches <- matches + 1
  }
}
probability <- matches / trials
return(probability)
}
birthday_same(10)
birthday_same(23)
M <- 1
prob <- 0
while (prob <= 0.5) {
prob <- birthday_same(M)
cat("M =", M, "Probability =", prob, "\n")
M <- M + 1
}
# OUTPUT (examples):
# birthday_same(10) -> approx 0.1152
# birthday_same(23) -> approx 0.4967
# Smallest M with probability > 0.5: 23


Q4
rain_given_clouds <- function(pC, pR, pC_given_R) {
prob <- (pC_given_R * pR) / pC
return(prob)
}
rain_given_clouds(0.4, 0.2, 0.85)
# OUTPUT:
# [1] 0.425


Q5
urn_prob <- function(red, total) {
p_U <- 1/3
p_R_given_U <- red / total
return(p_R_given_U * p_U)
}
denom <- urn_prob(8, 12) + urn_prob(6, 12) + urn_prob(5, 12)
```

```
cat("From I:", urn_prob(8, 12)/denom, "\n")
cat("From II:", urn_prob(6, 12)/denom, "\n")
cat("From III:", urn_prob(5, 12)/denom, "\n")
# OUTPUT:
# From I: 0.4210526
# From II: 0.3157895
# From III: 0.2631579


Q6
defective_car <- function() {
pA <- 2/5
pB <- 2/5
pC <- 1/5
dA <- 0.02
dB <- 0.03
dC <- 0.04
total_def <- dA*pA + dB*pB + dC*pC
p_A_given_def <- (dA*pA) / total_def
list(prob_defective = total_def, prob_A_given_def = p_A_given_def)
}
defective_car()
# OUTPUT:
# $prob_defective
# [1] 0.028
# $prob_A_given_def
# [1] 0.2857143


Q7
led_defect_prob <- function() {
pA <- 0.2; pB <- 0.3; pC <- 0.5
dA <- 0.01; dB <- 0.02; dC <- 0.03
total_def <- dA*pA + dB*pB + dC*pC
p_A_given_def <- (dA * pA) / total_def
p_B_given_def <- (dB * pB) / total_def
p_C_given_def <- (dC * pC) / total_def
list(
  prob_defective = total_def,
  prob_A_given_def = p_A_given_def,
  prob_B_given_def = p_B_given_def,
  prob_C_given_def = p_C_given_def
)
}
led_defect_prob()
# OUTPUT:
# $prob_defective
# [1] 0.023
# $prob_A_given_def
# [1] 0.08695652
# $prob_B_given_def
# [1] 0.2608696
# $prob_C_given_def
# [1] 0.6521739
```

ASSIGNMENT 3

Q1

```
x <- c(0,1,2,3,4)
y <- c(0.41,0.37,0.16,0.05,0.01)
EV1 <- sum(x^2 * y)
cat("EV1:", EV1, "\n")
EV2 <- sum((2*x + 3) * y)
cat("EV2:", EV2, "\n")
EV3 <- sum((2*x - 3) * y)
cat("EV3:", EV3, "\n")
EV4 <- sum((2*x - 3)^2 * y)
var <- EV4 - (EV3)^2
cat("variance :", var, "\n")
# OUTPUT:
# EV1: 1.62
# EV2: 4.76
# EV3: -1.24
# variance : 3.3824
```

Q2

```
x <- c(0, 1, 2, 3)
px <- c(0.1, 0.2, 0.2, 0.5)
y <- 24*x - 12*3 + 4*(3-x)  # as per given expression
EV <- sum(y * px)
V <- sum((y)^2 * px) - (EV)^2
cat("EV:", EV, "\n")
cat("variance :", V, "\n")
# OUTPUT:
# EV: 18
# variance : 436
```

Q3

```
X <- c(-3, -1, 0, 1, 2, 3, 5, 8)
F <- c(0.10, 0.30, 0.45, 0.5, 0.75, 0.90, 0.95, 1.0)
pmf <- c(F[1], diff(F))
cat("PMF :", pmf, "\n")
even <- X %% 2 == 0
P_even <- sum(pmf[even])
P_1to8 <- sum(pmf[X >= 1 & X <= 8])
P_gt0 <- sum(pmf[X > 0])
P_ge3_and_gt0 <- sum(pmf[X >= 3])
prob <- P_ge3_and_gt0 / P_gt0
list(P_even = P_even, P_1to8 = P_1to8, P_Xge3_given_Xgt0 = prob)
# OUTPUT:
# PMF : 0.10 0.20 0.15 0.05 0.25 0.15 0.05 0.05
# P_even = 0.45
# P_1to8 = 0.55
```

```
# P_Xge3_given_Xgt0 = 0.4545455


Q4
f <- function(t) { ifelse(t >= 0, 0.2 * exp(-0.2 * t), 0) }
mean_T <- integrate(function(t) t * f(t), 0, Inf)$value
mean_T2 <- integrate(function(t) t^2 * f(t), 0, Inf)$value
var_T <- mean_T2 - mean_T^2
mean_T
var_T
# For 2T - 3
mean_2T3 <- 2 * mean_T - 3
var_2T3 <- 4 * var_T
mean_2T3
var_2T3
# OUTPUT:
# mean_T = 5
# var_T = 25
# mean_2T3 = 7
# var_2T3 = 100


Q5
f <- function(x) { ifelse(x >= 0, 3 * exp(-3 * x), 0) }
mean_x <- integrate(function(x) x * f(x), 0, Inf)$value
mean_x2 <- integrate(function(x) x^2 * f(x), 0, Inf)$value
var_x <- mean_x2 - mean_x^2
mean_2X3 <- 2 * mean_x + 3
var_2X3 <- 4 * var_x
mean_x
var_x
mean_2X3
var_2X3
# OUTPUT:
# mean_x = 0.3333333
# var_x = 0.1111111
# mean_2X3 = 3.666667
# var_2X3 = 0.4444444


Q6
f <- function(x) { ifelse(x > 1 & x < 20, 0.5 * exp(-0.5 * x), 0) }
mean_x <- integrate(function(x) x * f(x), 1, 20)$value
mean_x2 <- integrate(function(x) x^2 * f(x), 1, 20)$value
var_x <- mean_x2 - mean_x^2
mean_x
var_x
# OUTPUT:
# mean_x = 1.818593
# var_x = 4.555462


Q7
```

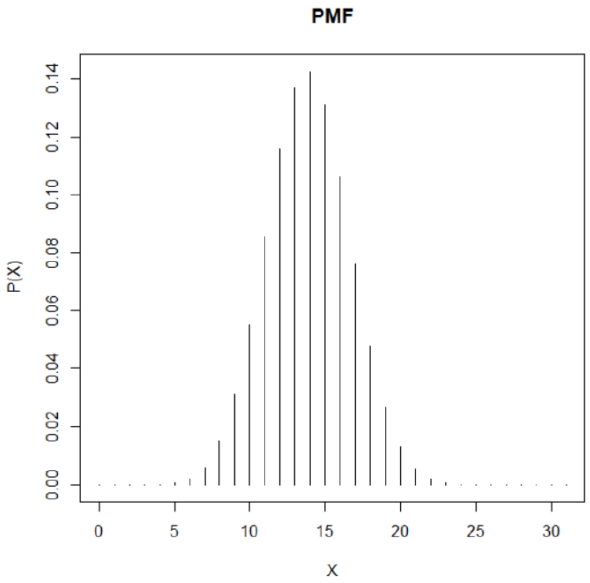
```
dice <- 1:6
outcomes <- expand.grid(dice, dice)
sums <- rowSums(outcomes)
mean_sums <- mean(sums)
var_sums <- var(sums)
mean_sums
var_sums
# OUTPUT:
# mean_sums = 7
# var_sums = 6


Q8
f <- function(x) { (3/4) * (1/4)^(x-1) }
x <- 1:10
y <- x^2
py <- f(x)
prob_Y_values <- py[1:5]
names(prob_Y_values) <- y[1:5]
P_Y3 <- f(3)
EY <- sum(y * py)
VarY <- sum((y - EY)^2 * py)
EY
VarY
prob_Y_values
P_Y3
# OUTPUT:
# EY = 2.222099
# VarY = 9.12025
# prob_Y_values for {1,4,9,16,25} = 0.75, 0.1875, 0.046875, 0.01171875, 0.002929688
# P_Y3 = 0.046875
```

ASSIGNMENT 4

```
Q1
n <- 12
p <- 1/6
p_a <- dbinom(8, n, p)
p_b <- pbinom(6, n, p)
p_c <- 1 - pbinom(6, n, p)
p_d <- pbinom(9, n, p) - pbinom(6, n, p)
p_e <- pbinom(8, n, p) - pbinom(7, n, p)
p_a; p_b; p_c; p_d; p_e
# OUTPUT:
# p_a = 0.0001421249
# p_b = 0.9987075
# p_c = 0.001292544
# p_d = 0.001291758
# p_e = 0.0001421249
```

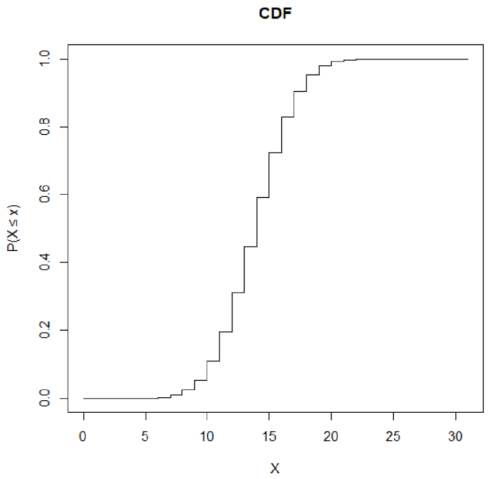
Q2

```
n <- 31
p <- 0.4477
pmf <- dbinom(0:n, n, p)
cdf <- pbinom(0:n, n, p)
mean_X <- n * p
var_X <- n * p * (1 - p)
sd_X <- sqrt(var_X)
mean_X; var_X; sd_X
# OUTPUT:
# mean_X = 13.8787
# var_X = 7.665206
# sd_X = 2.768611
```



Q3

```
n <- 2000
p <- 0.001
p_a <- dbinom(3, n, p)
p_b <- 1 - pbinom(2, n, p)
p_a; p_b
# OUTPUT:
# p_a = 0.1805373
# p_b = 0.3233236
```



Q4

```
n <- 100000
p <- 2e-5
lambda <- n * p
p_event <- 1 - ppois(4, lambda)
p_event
# OUTPUT:
# 0.05265302
```

Q5

```
# (a) λ = 5, Prob no car arrives
ppois(0, lambda = 5)
# OUTPUT: 0.006737947
```

```
# (b) For 10 hours λ = 50, P(48 ≤ X ≤ 50)
p_48_50 <- ppois(50, lambda = 50) - ppois(47, lambda = 50)
p_48_50
# OUTPUT: 0.1678485
```

Q6

```
p <- 0.6
max_k <- 100
probs_odd <- dgeom(seq(0, max_k, by = 2), p)
p_odd <- sum(probs_odd)
```

```
p_odd
# OUTPUT: 0.7142857


Q7
prob <- 0.05
# Prob 5th hit occurs on 10th throw (negative binomial)
p_prob <- dnbinom(5 - 1, size = 5, prob = prob)
p_prob
# OUTPUT: 1.781732e-05


Q8
r <- 2
p_def <- 0.05
# Prob at least 4 sampled to get 2 defective -> P(X ≥ 4)
p_event <- 1 - pnbinom(3 - 2, size = r, prob = p_def)
p_event
# OUTPUT: 0.99275
```

ASSIGNMENT – 5

Ques 1
If X is uniformly distributed over the interval [−2, 2] write an R program to find:
1. P(X < 0)
2. P(|X − 1| ≥ 1/2)

```
a <- -2
b <- 2

# 1. P(X < 0)
p1 <- punif(0, min = a, max = b)

# 2. P(|X - 1| >= 1/2)
# This is X <= 0.5 OR X >= 1.5
p2 <- punif(0.5, min = a, max = b) + (1 - punif(1.5, min = a, max = b))

p1
p2
```

Output: p1 = 0.5, p2 = 0.75

Ques 2

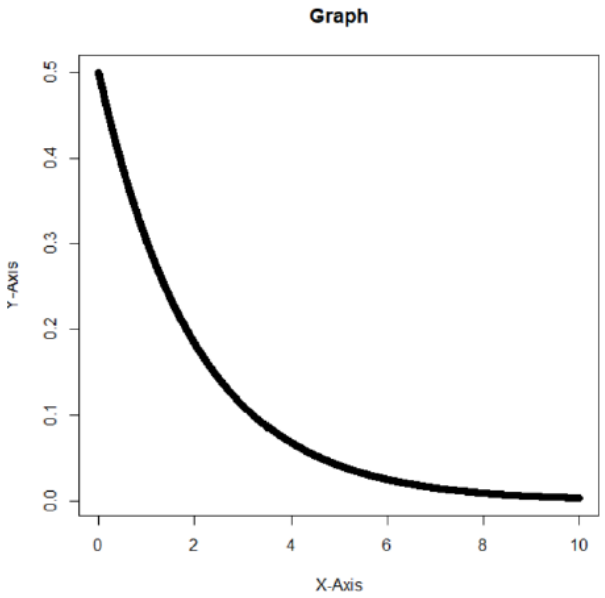
```
a <- 0
b <- 60

# (i) P(X > 35)
p1 <- 1 - punif(35, min = a, max = b)

# (ii) P(10 <= X <= 25)
p2 <- punif(25, min = a, max = b) - punif(10, min = a, max = b)

p1
p2
```

Output: p1 = 0.4166667, p2 = 0.25

Q3

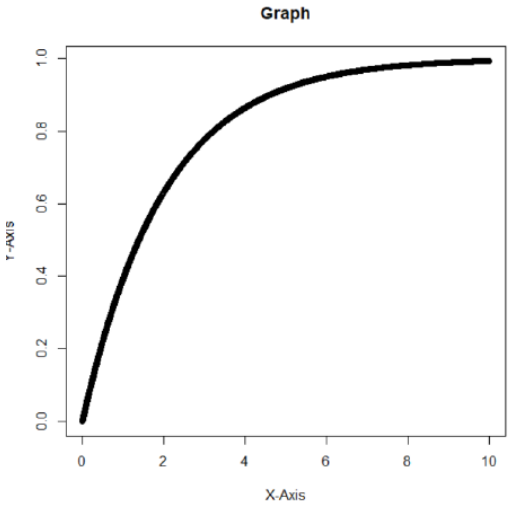
(i) dexp(4, 0.5)
Output: 0.06766764

(ii)
```
x <- seq(0, 10, 0.001)
y <- dexp(x, 0.5)
plot(x, y, xlab = "X-Axis", ylab = "Y-Axis", main = "Graph")
```

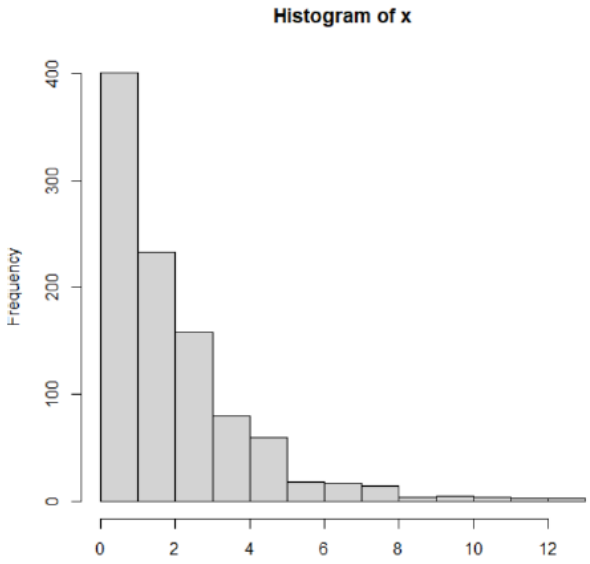


(iii) pexp(4, 0.5)
Output: 0.8646647

(iv)
```
x <- seq(0, 10, 0.001)
y <- pexp(x, 0.5)
plot(x, y, xlab = "X-Axis", ylab = "Y-Axis", main = "Graph")
```



(v)
```
x <- rexp(1000, 0.5)
hist(x)
```



Ques 4-a
```
1 - pexp(3, 2)
```

Ques 4-b
```
(1 - pexp(3, 2)) / (1 - pexp(2, 2))
```

Output: 0.002478752
0.1353353

Ques 5-a
```
300 * pnorm(57, 64.5, 5)
```

Ques 5-b
```
300 * (pnorm(72, 64.5, 5) - pnorm(57, 64.5, 5))
```

Output: 20.04216
259.9157

Ques 6-a
1 - pgamma(1, 2, , 1/3)

Ques 6-b
qgamma(0.7, 2, , 1/3)

Output: 0.1991483
0.8130722

Ques 7
```
q1 <- 30
q2 <- 62
z1 <- qnorm(0.1)
z2 <- qnorm(0.97)
s <- (q2 - q1) / (z2 - z1)
s
mean <- q1 - s * z1
mean
```

Output: 10.11907
42.96811

ASSIGNMENT – 6

Q1

```
# Define x and y values
x <- 0:2
y <- 0:2

# Create joint pmf matrix
pmf <- outer(x, y, function(a, b) (2*a + b) / 27)
pmf

## (i) Display joint pmf
print("Joint PMF matrix:")
pmf

## (ii) Check if it is a valid pmf (sum should be 1)
total_sum <- sum(pmf)
total_sum
```

```
## (iii) Marginal distribution pX(x)
pX <- apply(pmf, 1, sum)
pX

## (iv) Marginal distribution pY(y)
pY <- apply(pmf, 2, sum)
pY

## (v) Conditional probability P(X=0 | Y=1)
P_X0_given_Y1 <- pmf[1, 2] / pY[2]   # pmf[x=0,y=1] / pY(1)
P_X0_given_Y1

## (vi) Expectations, variances, covariance, correlation

# Expected X
E_X <- sum(x * pX)

# Expected Y
E_Y <- sum(y * pY)

# Expected XY
E_XY <- sum(outer(x, y, "*") * pmf)

# Var(X)
E_X2 <- sum((x^2) * pX)
Var_X <- E_X2 - (E_X)^2

# Var(Y)
E_Y2 <- sum((y^2) * pY)
Var_Y <- E_Y2 - (E_Y)^2

# Cov(X,Y)
Cov_XY <- E_XY - (E_X * E_Y)

# Correlation coefficient
rho <- Cov_XY / sqrt(Var_X * Var_Y)

# Display all
E_X; E_Y; E_XY
Var_X; Var_Y
Cov_XY
rho
```

Q2

```
# Joint PDF
f <- function(x, y) {
  ifelse(x >= 0 & x <= 1 & y >= 0 & y <= 1,
         (2/5) * (x + 4*y),
         0)
}

# (i) Check if valid PDF: ∫0^1 ∫0^1 f(x,y) dx dy
total_integral <- integral2(f, xmin = 0, xmax = 1, ymin = 0, ymax = 1)
total_integral


# (ii) Marginal f_X(x) = ∫0^1 f(x,y) dy   evaluated at x = 1
fX <- function(x) {
  sapply(x, function(xx)
    integrate(function(yy) f(xx, yy), 0, 1)$value)
}

fX_1 <- fX(1)
fX_1


# (iii) Marginal f_Y(y) = ∫0^1 f(x,y) dx   evaluated at y = 0
fY <- function(y) {
  sapply(y, function(yy)
    integrate(function(xx) f(xx, yy), 0, 1)$value)
}

fY_0 <- fY(0)
fY_0


# (iv) Expected value E[g(X,Y)] = E(XY)
E_XY <- integral2(
  function(x, y) x * y * f(x, y),
  xmin = 0, xmax = 1,
  ymin = 0, ymax = 1
)

E_XY
```

ASSIGNMENT – 7

Ques 1

The average height of residents is 168 cm with σ = 3.9 cm.
Doctor measured 36 individuals and found mean = 169.5 cm.

(a) State Null and Alternate Hypothesis
(b) At 95% CI, test if Null can be rejected.

```
h0 = 168
n = 36
s = 3.9
m = 169.5
cl = 0.95


z = (m - h0) / (s / sqrt(n))
p = 2 * (1 - pnorm(abs(z)))


if(p > 0.05) cat("accepted")
else cat("rejected")
```

Output: rejected

Ques 2
Average height = 168, σ = 3.9, sample size = 25, mean = 169.5

```
t = (m - h0) / (s / sqrt(n))
p = 2 * (1 - pt(abs(t), n - 1))
```

Decision: rejected

Ques 3
```
t.test(x, y, alternative = "greater", mu = 0, paired = TRUE, var.equal = FALSE, cl = 0.95)
```

Output:
Paired t-test
t = -2.2597, df = 9, p-value = 0.9749
95% CI: -7.244834 to Inf
mean difference = -4